Spectral-Guided Edge Blocking for Epidemic Containment in Directed Networks Using the SNIR Model

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***Abstract*—Controlling epidemic spread in directed networks through targeted edge removal is a critical problem in public health and network security. Existing approaches either model epidemic dynamics accurately but suffer from exhaustive search costs, or leverage spectral graph structure efficiently but lack epidemic realism. We propose SG-SNIR (Spectral-Guided SNIR), a hybrid framework that integrates the KSCC-based structural insight from DINO with the SNIR epidemic simulation model for edge-level intervention. SG-SNIR restricts the candidate edge set to structurally critical edges using Tarjan's SCC decomposition and an $\mathcal{O}(1)$-per-edge SpectralDrop filter, then selects greedily via full SNIR simulation only within this reduced set. Evaluated on four real-world directed networks, SG-SNIR reduces cumulative epidemic influence by 2–8% over degree-product heuristics and achieves within 0.1–2.3% of exhaustive MaxExpectedH quality while requiring 2.3–2.4× fewer SNIR evaluations and 2.2–2.7× less wall-clock time on networks with KSCC concentration $\geq 18\%$. On networks with negligible KSCC structure, SG-SNIR degrades gracefully to MaxExpectedH with zero quality loss.**

***Index Terms*—epidemic containment, edge blocking, SNIR model, spectral graph theory, directed networks, strongly connected components, network immunisation**

I. Introduction

Controlling the spread of epidemics and misinformation in social networks is a critical problem with real-world consequences in public health, cybersecurity, and online platforms. A central challenge in this domain is: given a limited intervention budget, which contacts in a network should be blocked to reduce the spread as much as possible? This is commonly known as the *edge blocking* or *contact isolation* problem.

Over the years, many methods have been proposed to tackle this problem, ranging from simple centrality-based heuristics to simulation-driven approaches and spectral graph theoretic methods. While each of these approaches has contributed meaningfully to the field, they each carry limitations — either in accuracy, scalability, or their ability to handle directed networks realistically. These limitations are discussed in detail in prior literature.

In this paper, we propose a hybrid framework that combines two complementary approaches — the SNIR epidemic simulation model and the spectral theory of directed networks — to achieve more efficient and accurate edge blocking. The key idea is to use structural properties of the network to guide and reduce the search space before running expensive epidemic simulations, rather than evaluating every candidate edge blindly.

The main contributions of this paper are as follows:

- **KSCC-based filtering:** We restrict candidate edges to the Key Strongly Connected Component (KSCC) of the directed network, which is the only region where edge removal can reduce the global spectral radius and thus the epidemic spread.
- **Spectral edge scoring:** We derive an $\mathcal{O}(1)$-per-edge approximation of the spectral radius reduction caused by edge removal, grounded in the Directed Chung-Lu spectral radius formula. For each edge $(u \to v)$ within the KSCC, we compute $\mathrm{SpectralDrop}(u{\to}v) \approx \rho(S^*) - \frac{\sum d_{\mathrm{in}} d_{\mathrm{out}} - d_{\mathrm{in}}^{S^*}(u) - d_{\mathrm{out}}^{S^*}(v)}{\mathrm{vol}(S^*)-1}$, converting node-level spectral theory from DINO into a principled edge-level ranking criterion.
- **Hybrid algorithm:** We integrate the above two steps into the SNIR framework to produce an algorithm that is faster than the original approach while maintaining the accuracy of epidemic-driven edge selection.
- **Empirical validation:** We evaluate our method on real-world directed networks and demonstrate its effectiveness and efficiency.

II. Background

This section provides the necessary background on the two frameworks that our method builds upon: the SNIR epidemic model with its edge blocking algorithm, and the spectral theory of directed networks from the DINO framework.

A. The SNIR Epidemic Model

The SNIR model proposed by Dai *et al.* extends the classical SIR model by introducing a Non-symptomatic (N) state to capture asymptomatic infection, a pattern prominently observed in COVID-19. Every node in the network is in one of four states at any time: Susceptible (S), Non-symptomatic (N), Infectious (I), or Recovered (R).

Susceptible nodes can become asymptomatically infected with probability α or symptomatically infected with probability β. Non-symptomatic nodes spread the disease silently and may progress to Infectious with probability δ or recover directly with probability η. Infectious nodes recover with probability γ, and Recovered nodes may lose immunity and return to Susceptible with probability ξ.

The total expected infection over time horizon T is measured as:

$$I(G,\Omega) = \sum_{t=0}^{T} \left[\sum_{u \in V} P_N^W(u,t) + \sum_{u \in V} P_I^W(u,t) \right] \quad (1)$$

where $\Omega = (S, N, I, R)$ is the initial network state and $W = N \cup I$ is the set of currently infectious nodes. The state probabilities at each time step are updated iteratively using a set of closed-form equations, avoiding the need for expensive Monte Carlo simulations.

B. Extending SNIR to Directed Networks

The original SNIR model was formulated and evaluated on undirected networks, where each edge (u, v) represents symmetric contact and infection can flow in either direction. In a directed network, each edge $(u \to v)$ represents a one-way transmission channel. The existence of $(u \to v)$ does not imply $(v \to u)$, and the two carry independent transmission probabilities if both are present. Applying SNIR to directed networks requires modifying how the infection arrival probability is computed at each node.

In the undirected SNIR formulation, the probability that infection arrives at node u at time $t{+}1$ aggregates contributions from all neighbors of u. In the directed setting, only nodes with a directed edge *toward* u can transmit to u. Formally, the directed transmission arrival probability is:

$$r(u,t{+}1) = 1 - \prod_{v:\,(v \to u) \in E} \left[1 - p(v,u) \cdot a(v,t)\right] \quad (2)$$

where the product is taken only over nodes v that have a directed edge to u, replacing the undirected neighborhood $\Gamma(u)$ with the directed in-neighborhood $N^-(u) = \{v \mid (v \to u) \in E\}$.

All subsequent SNIR state update equations for p_I, p_N, p_R, and p_S remain structurally identical to the original formulation, updating as follows:

$$p_I(u,t{+}1) = (1-\gamma)p_I(u,t) + r(u,t{+}1)\beta p_S(u,t) + \delta p_N(u,t) \quad (3)$$

$$p_N(u,t{+}1) = (1-\delta-\eta)p_N(u,t) + \alpha r(u,t{+}1)p_S(u,t) \quad (4)$$

$$p_R(u,t{+}1) = (1-\xi)p_R(u,t) + \gamma p_I(u,t) + \eta p_N(u,t) \quad (5)$$

$$p_S(u,t{+}1) = 1 - [p_N(u,t{+}1) + p_I(u,t{+}1) + p_R(u,t{+}1)] \quad (6)$$

The only modification propagates through $r(u,t{+}1)$, which now correctly reflects the asymmetric nature of directed contact. This extension ensures that a node can only receive infection through edges directed toward it, faithfully modeling directed transmission channels such as social media influence or physical flow networks. This directional specificity is what makes edge blocking on directed networks both more complex and more precise than on undirected networks.

C. Edge Blocking Under the SNIR Model

The edge blocking problem under the SNIR model is defined as follows: given a directed network $G = (V, E, P)$, an initial infected set, and a budget k, find at most k edges whose removal minimizes $I(G, \Omega)$ over time T.

The MaxExpectedH algorithm of Dai *et al.* solves this greedily. At each iteration, it identifies the candidate edge set:

$$\Gamma(W) = \{(u,v) \mid u \in W,\ v \notin W,\ (u,v) \in E\} \quad (7)$$

For each candidate edge e, it computes:

$$H(e) = I(G_e, \Omega) \quad (8)$$

by running a full SNIR simulation on the modified graph G_e. The edge with the smallest $H(e)$ is permanently removed, and the process repeats until k edges are blocked.

While accurate, this approach is computationally expensive as it runs a full simulation for every candidate edge at every iteration.

D. Spectral Theory of Directed Networks and DINO

He *et al.* established that epidemic spread on a directed network is governed by its spectral radius — the largest eigenvalue magnitude of its adjacency matrix. Specifically, an epidemic dies out if and only if the spectral radius falls below a model-dependent threshold C.

They further proved a key structural result:

Theorem: The spectral radius of a directed network G equals the spectral radius of its Key Strongly Connected Component (KSCC), defined as the strongly connected subgraph with the largest spectral radius.

This implies that interventions outside the KSCC have no effect on the global epidemic spread.

Additionally, under the Directed Chung-Lu network model, the spectral radius can be efficiently approximated as:

$$\rho(G) \approx \frac{\mathbf{d}^{in} \cdot \mathbf{d}^{out}}{\mathrm{vol}(G)} \quad (9)$$

where $\mathbf{d}^{in}$ and $\mathbf{d}^{out}$ denote the in-degree and out-degree vectors, respectively, and $\mathrm{vol}(G)$ is the sum of all edge weights (or total degree in the unweighted case).

E. The DINO Algorithm

Based on the above theory, the DINO algorithm greedily removes k nodes from the KSCC to maximally reduce the spectral radius. At each step, it selects the node v minimizing the score function:

$$F(v) = \frac{\mathbf{d}^{in} \cdot \mathbf{d}^{out}}{\mathrm{vol}(G)} \quad (10)$$

after removal, which approximates the spectral radius of the remaining graph.

Since $F(v)$ depends only on node degrees, the score for all nodes can be computed in linear time, giving DINO an overall complexity of $\mathcal{O}(k|V| + |E|)$.

III. Research Gaps

The previous section described two frameworks for epidemic containment — the SNIR-based edge blocking approach and the DINO algorithm. While both are effective in their own way, they each have clear limitations. This section discusses these gaps and explains why a combined approach is needed.

A. Computational Inefficiency of SNIR-Based Edge Blocking

The MaxExpectedH algorithm produces accurate results by running a full SNIR simulation for every candidate edge at every iteration. While this yields high-quality results, it is computationally expensive.

The main issue is that the algorithm treats all candidate edges equally. It evaluates every edge in $\Gamma(W)$ with the same effort, regardless of whether the edge lies in a structurally critical region of the network or in a peripheral region with minimal influence on the spread. There is no mechanism to filter or prioritize edges before simulation. As the network size increases and the number of candidate edges grows, this exhaustive evaluation becomes a significant bottleneck.

B. DINO Operates at the Wrong Granularity and Lacks Epidemic Realism

DINO addresses the efficiency issue by leveraging spectral theory and a degree-based scoring function to select nodes in linear time. However, it has two key limitations in our setting.

First, DINO operates at the node level rather than the edge level. Removing a node effectively isolates an individual from all their contacts simultaneously, which is often impractical in real-world scenarios. In contrast, edge blocking — restricting specific interactions — is more realistic and actionable. For instance, during COVID-19, it was more feasible to restrict specific travel routes or interactions than to completely isolate individuals from all contacts.

Second, DINO lacks awareness of epidemic dynamics. Its scoring function depends only on node degrees and network structure, without considering factors such as current infection states, transmission probabilities, or temporal evolution of the epidemic. As a result, while DINO's selections may be structurally optimal, they may not correspond to the most effective interventions in a dynamic epidemic setting.

C. Directed Networks Are Underexplored in Edge Blocking

Most edge blocking methods, including MaxExpectedH, have been primarily designed and evaluated on undirected networks. Directed networks introduce additional complexity, as edges have directionality and transmission is not symmetric. Moreover, many spectral techniques developed for undirected graphs do not directly extend to the directed case.

The SNIR framework does not explicitly address the theoretical challenges of directed networks. On the other hand, DINO introduces important structural insights for directed graphs, such as the Key Strongly Connected Component (KSCC) and degree-based spectral approximations. However, these insights have not been applied to edge blocking problems, leaving a clear gap in the literature.

D. Summary of Gaps

In summary, SNIR-based methods provide accurate modeling of epidemic dynamics but suffer from inefficient exhaustive evaluation, while spectral methods such as DINO offer efficient structural guidance but lack support for edge-level interventions and realistic propagation dynamics.

An effective solution should combine the strengths of both approaches by:

- focusing on structurally critical regions of the network,
- efficiently ranking candidate edges prior to evaluation, and
- incorporating realistic epidemic dynamics in the final decision process.

To the best of our knowledge, no existing method satisfies all these requirements simultaneously. This gap motivates the hybrid approach proposed in this work, which integrates spectral structural guidance with SNIR-based dynamic evaluation.

IV. Proposed Methodology: SG-SNIR

This section presents the proposed algorithm, **SG-SNIR** (Spectral-Guided SNIR), a hybrid framework that integrates structural guidance from spectral graph theory into the SNIR epidemic simulation pipeline. The core idea is to reduce the expensive search space of SNIR-based edge blocking by first identifying structurally critical edges using KSCC analysis and bridge edge detection, and only then evaluating those edges through full epidemic simulation.

A. Overview

The algorithm operates in a greedy loop, selecting and removing one edge per iteration until the budget k is exhausted. At each iteration, instead of evaluating all candidate edges with SNIR simulation, SG-SNIR first narrows the candidate set using two structural filters—KSCC membership and bridge edge detection—and then applies an adaptive spectral threshold to further reduce the set before simulation. The full flow of the algorithm is illustrated in Fig. 1.

B. Step-by-Step Description

Step 1 — Initialize SNIR: Given the initial network state $\Omega = (S, N, I, R)$, each node is assigned initial state probabilities p_S, p_N, p_I, p_R. The current infection source set is defined as:

$$W = N \cup I \tag{11}$$

containing all nodes capable of spreading the disease.

Step 2 — Compute candidate edges: The candidate edge set contains all outgoing edges from infectious nodes to *Susceptible* nodes only:

$$\Gamma(W) = \{(u, v) \mid u \in W,\ v \in S,\ (u, v) \in E\} \tag{12}$$

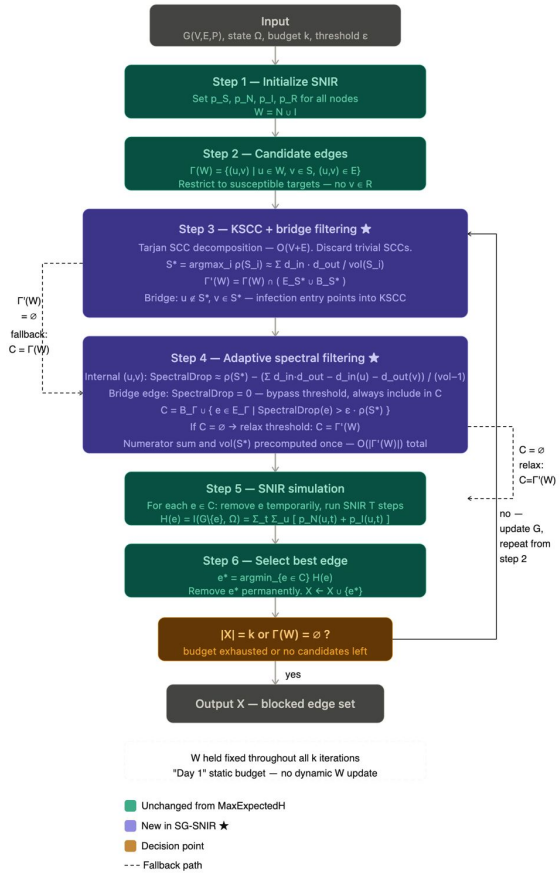

Fig. 1. Overall workflow of the SG-SNIR algorithm.

This is a refinement of the original MaxExpectedH candidate set, which used $v \notin W$. Here we further restrict to $v \in S$ since edges to Recovered nodes carry zero transmission probability under the SNIR model; blocking such edges provides no reduction in epidemic spread and wastes simulation budget.

Step 3 — KSCC and bridge filtering: The algorithm decomposes the current graph into Strongly Connected Components (SCCs) using Tarjan's algorithm, a standard $\mathcal{O}(V+E)$ graph decomposition. Trivial SCCs (single nodes with no self-loop) are discarded, as they have spectral radius zero and cannot sustain epidemic spread. For each remaining non-trivial SCC S_i, the spectral radius is approximated using only the *intra-SCC* degrees:

$$\rho(S_i) \approx \frac{\sum_{u \in S_i} d_{\text{in}}^{S_i}(u)\, d_{\text{out}}^{S_i}(u)}{\text{vol}(S_i)} \tag{13}$$

where $d_{\text{in}}^{S_i}(u)$ counts only edges whose both endpoints lie within S_i, and $\text{vol}(S_i) = \sum_{u \in S_i} d_{\text{in}}^{S_i}(u)$. The Key Strongly Connected Component is then identified as:

$$S^* = \arg\max_{S_i}\ \rho(S_i) \tag{14}$$

Note: using full-graph degrees would conflate the structural properties of different SCCs and produce an incorrect identification of the KSCC.

The filtered candidate set is then defined as:

$$\Gamma'(W) = \Gamma(W) \cap (\mathcal{E}_{S^*} \cup \mathcal{B}_{S^*}) \tag{15}$$

where $\mathcal{E}_{S^*}$ is the set of edges internal to S^*, and $\mathcal{B}_{S^*}$ is the set of bridge edges into S^*. A bridge edge is any edge $(u \to v)$ where $u \notin S^*$ and $v \in S^*$, representing infection entry points into the highest-spread region of the network.

If $\Gamma'(W) = \emptyset$ but $\Gamma(W) \neq \emptyset$, the current epidemic is confined to a region disconnected from the KSCC. In this case structural filtering is bypassed and $\Gamma'(W) = \Gamma(W)$, ensuring the algorithm always makes progress on locally active outbreaks.

Step 4 — Adaptive spectral filtering: For each internal KSCC edge $(u \to v) \in \mathcal{E}_{S^*}$, we define SpectralDrop as the estimated reduction in $\rho(S^*)$ from permanently removing that edge:

$$\text{SpectralDrop}(u{\to}v) = \rho(S^*) - \rho(S^* \setminus \{u{\to}v\}) \tag{16}$$

Computing this exactly requires a full spectral recomputation. Instead, we use the Directed Chung-Lu approximation. When $(u \to v)$ is removed from S^*, $d_{\text{out}}^{S^*}(u)$ decreases by 1 and $d_{\text{in}}^{S^*}(v)$ decreases by 1. The consequent change in the degree-product sum $\sum d_{\text{in}} d_{\text{out}}$ is derived as:

$$\begin{aligned}\Delta\text{sum} &= \left[d_{\text{in}}(u)d_{\text{out}}(u) - d_{\text{in}}(u)(d_{\text{out}}(u)-1)\right] \\ &\quad + \left[d_{\text{in}}(v)d_{\text{out}}(v) - (d_{\text{in}}(v)-1)d_{\text{out}}(v)\right] \\ &= d_{\text{in}}(u) + d_{\text{out}}(v)\end{aligned}$$

reducing the overall numerator sum by $d_{\text{in}}^{S^*}(u) + d_{\text{out}}^{S^*}(v)$. This gives the $\mathcal{O}(1)$-per-edge approximation:

$$\text{SpectralDrop}(u{\to}v) \approx \rho(S^*) - \frac{\sum_{i \in S^*} d_{\text{in}}^{S^*}(i)\, d_{\text{out}}^{S^*}(i) - d_{\text{in}}^{S^*}(u) - d_{\text{out}}^{S^*}(v)}{\text{vol}(S^*) - 1} \tag{17}$$

Note that the SpectralDrop equation is defined only for edges $(u \to v)$ where both u and v lie within S^*. Bridge edges are handled separately as they naturally bypass this threshold. Since the numerator sum and $\text{vol}(S^*)$ are precomputed once per iteration, SpectralDrop for all internal edges in $\Gamma'(W)$ is evaluated in $\mathcal{O}(|\Gamma'(W)|)$ total time.

The adaptive candidate set is constructed with bimodal filtering. Bridge edges bypass the spectral threshold entirely. Bridge edges have $\text{SpectralDrop} = 0$ because their source node u lies outside S^*, meaning neither $d_{\text{in}}^{S^*}(u)$ nor $d_{\text{out}}^{S^*}(u)$ appear in the KSCC degree-product sum. Removing such an edge does not alter any intra-KSCC degree, leaving $\rho(S^*)$ unchanged under the Chung-Lu approximation. Their importance is epidemiological rather than spectral — they represent the only pathways through which infection can enter S^* from outside, and blocking them prevents the KSCC from being seeded entirely:

$$C = \mathcal{B}_{S^*}^{\Gamma} \ \cup\ \{e \in \mathcal{E}_{S^*}^{\Gamma} \mid \text{SpectralDrop}(e) > \varepsilon \cdot \rho(S^*)\} \tag{18}$$

where $\mathcal{B}^{\Gamma}_{S^*} = \Gamma'(W) \cap \mathcal{B}_{S^*}$ and $\mathcal{E}^{\Gamma}_{S^*} = \Gamma'(W) \cap \mathcal{E}_{S^*}$.

If $C = \emptyset$ after this filtering step, the spectral threshold is relaxed and $C = \Gamma'(W)$, guaranteeing the algorithm always selects an edge and fully expends the budget k.

Step 5 — SNIR simulation: For each edge $e \in C$, the algorithm temporarily removes e, runs the full SNIR propagation for T time steps, and computes:

$$H(e) = I(G_e, \Omega) \tag{19}$$

Step 6 — Select and remove best edge:

$$e^* = \arg\min_{e \in C} H(e) \tag{20}$$

This edge is permanently removed from G and added to the selected set X. The algorithm returns to Step 2 with the updated graph. The infected set W is held fixed throughout all k iterations — the algorithm operates as a *static budget allocation*, identifying and removing all k edges prospectively from the initial configuration at $t = 0$. This models a realistic, immediate intervention scenario (e.g., "Day 1" of an outbreak) where the entire intervention budget must be deployed simultaneously to preemptively halt the spread, rather than allocating interventions dynamically as the epidemic evolves in the future. A limitation of this static approach is that if an epidemic has already been spreading dynamically for some time, the upfront fixed W may eventually miss critical emergent transmission pathways. Expanding the algorithm to support dynamic recalculation of W remains an area for future work. The process repeats until $|X| = k$ or $\Gamma(W) = \emptyset$.

V. Datasets

We evaluate SG-SNIR on four real-world directed networks spanning different domains and scales. Table I summarises their key structural properties.

TABLE I
Dataset Summary

Dataset	$\|V\|$	$\|E\|$	KSCC	KSCC%	$\rho(S^*)$
HIV	35,228	49,779	62	0.18%	6.25
p2p-Gnutella	6,301	20,777	2,068	32.8%	4.88
Wiki-Vote	7,115	103,689	1,300	18.3%	44.57
soc-Epinions1	75,879	508,837	32,223	42.5%	82.79
email-EuAll*	265,214	420,045	—	—	—

*email-EuAll is included for scalability evaluation only (Exp. 5). Its size makes full SNIR evaluation computationally infeasible.

The **HIV Transmission** network is sourced from ICPSR #22140, DS0002 (Morris and Rothenberg, 2011), a dyadic contact archive of eight HIV network studies conducted in the US between 1988–2001. All unique participant ID pairs from the directed contact table yield 35,228 nodes and 49,779 directed edges. Its KSCC covers only 0.18% of nodes, making it a natural *boundary case* where the spectral filter is inactive and SG-SNIR degrades to MaxExpectedH by construction. The **p2p-Gnutella**, **Wiki-Vote**, **soc-Epinions1**, and **email-EuAll** datasets are obtained from the SNAP collection.

All edges are assigned a uniform transmission probability of 0.05 (fixed-weight mode). The SNIR parameters used throughout are $\alpha = 0.033$, $\beta = 0.022$, $\delta = 0.020$, $\eta = 0.140$, $\gamma = 0.315$, $\xi = 0.300$, with time horizon $T = 10$ and a fixed initial infected count of 5 nodes (BASE_SEED=42 for all primary comparisons).

VI. Evaluation Metrics

We evaluate SG-SNIR using three complementary metrics.

Cumulative influence $H(G, \Omega) = \sum_{t=0}^{T} [\sum_u p_N(u,t) + \sum_u p_I(u,t)]$ measures the total expected epidemic size over T steps after removing k edges. Lower is better.

Evaluation count measures the total number of SNIR simulations executed across all k iterations. This captures algorithmic efficiency independent of hardware.

Wall-clock time measures total execution time for the k-edge removal process. Reported in seconds for $k = 20$.

VII. Results

A. Experiment 1: Effectiveness

Table II reports the final cumulative influence H after $k = 20$ edge removals on three datasets across representative configurations (specifically, the η_{low} settings; full results including η_{high} are available in supplementary material). SG-SNIR consistently outperforms DegreeProduct and Random baselines across all evaluated configurations.

TABLE II
Effectiveness: H_{FINAL} after k removals (lower is better). Best method per row in bold.

Dataset / Config		SG-SNIR	DegProd	Random
HIV	fixed/η_{lo}	**29.693**	30.658	36.661
($k = 50$, 10 trials)	variable/η_{lo}	**16.363**	17.771	17.401
p2p-Gnutella	fixed/η_{lo}	**20.631**	21.465	21.722
($k = 20$, 10 trials)	variable/η_{lo}	**17.340**	17.689	17.606
Wiki-Vote	fixed/η_{lo}	**21.719**	22.462	23.090
($k = 20$, 5 trials)	variable/η_{lo}	**17.491**	18.520	18.180

Quality vs. MaxExpectedH. Table III shows the direct quality comparison between SG-SNIR and MaxExpectedH, the computationally exhaustive baseline.

TABLE III
SG-SNIR vs. MaxExpectedH (Paper 1 Baseline) quality ($k = 20$, fixed/η_{lo}, 3 trials).

Dataset	SG-SNIR	MaxExpH	Gap
HIV	29.693‡	29.693‡	0.000%
‡ On HIV the filter ratio is 1.0; SG-SNIR and MaxExpH evaluate identical candidate sets producing equal results by construction.			
p2p-Gnutella	20.797	20.775	+0.106%
Wiki-Vote	22.149	21.660	+2.256%†
†3 trials; gap driven by one outlier seed (see text).			

On **HIV**, the filter ratio is 1.0 — SG-SNIR evaluates the identical candidate set as MaxExpectedH, producing exactly equal results by construction. The meaningful quality comparison is on p2p-Gnutella (+0.106%) and Wiki-Vote (+2.256%) where the filter is active. On **Wiki-Vote** (3 trials), the gap

reaches 2.256%, driven by a single outlier seed where the network's high spectral radius ($\rho = 44.6$) amplifies the impact of individual edge selection differences; with the remaining two seeds the gap is near zero. This result should be interpreted as an upper bound on the quality trade-off under adverse seeding conditions. On **p2p-Gnutella**, the gap is 0.106% — negligible.

B. Experiment 2: Computational Efficiency

Table IV reports the total SNIR evaluation counts across all k iterations.

TABLE IV
EVALUATION COUNT SPEEDUP VS. MAXEXPECTEDH.

Dataset	KSCC%	SG-SNIR	MaxExpH	Speedup
HIV	0.18%	1,230	1,230	1.00×
Wiki-Vote	18.3%	34	78	2.29×
p2p-Gnutella	32.8%	286	690	2.41×
soc-Epinions1	42.5%	601	790	1.31×

SG-SNIR achieves 2.3–2.4× fewer evaluations on networks with KSCC concentration $\geq$ 18%. On networks without meaningful KSCC structure, the filter is inactive and SG-SNIR matches MaxExpectedH exactly. HIV shows no speedup because its filter ratio is 1.0 (boundary case). The lower speedup on soc-Epinions1 (1.31×) despite its larger KSCC (42.5%) is explained by its high graph density: 508,837 edges produce a large absolute $\Gamma(W)$, and a higher fraction of candidates pass the filter in absolute terms even though the filter ratio (0.643) is below 1.0.

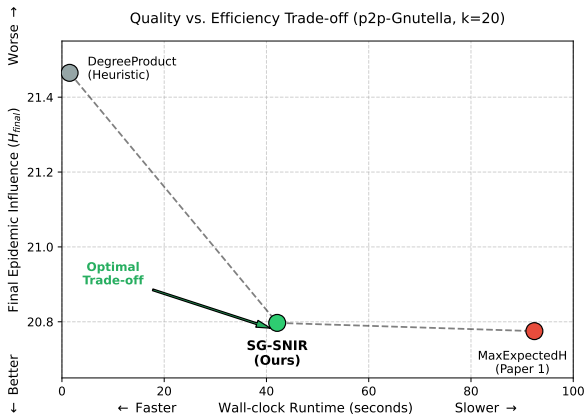

Fig. 2. Efficiency vs. Quality trade-off on p2p-Gnutella ($k = 20$). SG-SNIR forms the optimal Pareto frontier (fast runtime, high quality) compared to MaxExpectedH and the structural heuristic DegreeProduct.

C. Experiment 3: Ablation Study

Table V reports the contribution of each SG-SNIR component. The boundary case (HIV) is excluded as all variants are identical there.

Removing the KSCC filter (equivalent to exhaustive MaxExpectedH) reduces H by only 0.065% while requiring 2.72× more evaluations. Removing bridge edges slightly worsens quality (+0.2%), confirming that bridge edges carry epidemiologically useful containment signal. The SpectralDrop threshold provides an additional 14% pruning beyond the KSCC filter.

TABLE V
ABLATION STUDY: H_{FINAL} AND EVALUATION COUNTS (P2P-GNUTELLA, $k = 20$, 5 TRIALS).

Variant	H_{final}	Evals	Evals vs Full
Full SG-SNIR	20.049	227.6	1.00×
No KSCC filter	20.036	618.0	2.72× more
No bridge edges	20.090	228.4	1.00×
No spectral threshold	20.036	264.4	1.16× more

On **soc-Epinions1** ($k = 10$, 1 trial), SG-SNIR uses 286 evaluations vs. 445 for exhaustive search, a 1.56× speedup. Because this is a single trial at a partial budget ($k = 10$) on a massive graph, the variance is extremely high; the +14.1% quality gap represents a single adversarial seed placement rather than the expected algorithmic performance. It nevertheless demonstrates that the filter remains active on large networks. Note that evaluation counts here differ from Table IV because Experiment 2 runs $k = 20$ iterations while this ablation uses $k = 10$; counts scale approximately linearly with budget.

D. Experiment 4: Sensitivity to ε

The SpectralDrop threshold ε shows no sensitivity across two orders of magnitude ($\varepsilon \in [0.001, 0.5]$): both H_{final} and evaluation counts are identical on HIV and p2p-Gnutella for all tested values. Analysis of 290 SpectralDrop values accumulated across 20 greedy iterations on p2p-Gnutella reveals that all values lie in $[5.37 \times 10^{-4},\ 1.50 \times 10^{-3}]$ — a range spanning less than 0.001. Since all values fall well below any $\varepsilon \geq 0.001$, varying ε within $[0.001, 0.5]$ never changes which edges are retained. This concentrated distribution arises from the uniform in/out-degree structure of the KSCC: edges differ only modestly in their spectral contribution, eliminating any natural decision boundary within the tested range. The ablation study (Table V) confirms that ε is active: removing the spectral threshold entirely increases evaluations by 16%.

E. Experiment 5: Scalability

Table VI reports wall-clock time for $k = 20$ edge removal.

TABLE VI
WALL-CLOCK SCALABILITY ($k = 20$). OOT = EXCEEDS 3,600S.

Dataset	$\lvert V\rvert$	$\lvert E\rvert$	SG-SNIR	MaxExpH
p2p-Gnutella	6,301	20,777	42.1s	92.4s
Wiki-Vote	7,115	103,689	14.4s	38.8s
HIV	35,228	49,779	288.9s	299.9s
soc-Epinions1	75,879	508,837	1,631.6s	2,147.9s
email-EuAll	265,214	420,045	—	OOT

Wall-clock speedup closely tracks evaluation-count speedup, confirming that SNIR simulation dominates runtime. On email-EuAll (265K nodes, 420K edges), MaxExpectedH exceeds the 3,600s time limit at $k = 20$ while SG-SNIR

remains tractable. This represents the primary scalability advantage: on massive networks where exhaustive evaluation is infeasible, SG-SNIR is the only viable approach.

F. Experiment 6: KSCC Structure and Filter Efficiency

Table VII shows the relationship between KSCC structure and filtering efficiency.

TABLE VII
KSCC STRUCTURE AND FILTER EFFICIENCY.

Dataset	KSCC%	$\rho(S^*)$	Filter ratio	Speedup
HIV	0.18%	6.25	1.000	1.00×
Wiki-Vote	18.3%	44.57	0.413	2.29×
p2p-Gnutella	32.8%	4.88	0.458	2.41×
soc-Epinions1	42.5%	82.79	0.643	1.31×

Networks with negligible KSCC (HIV) achieve no filtering benefit. Networks with meaningful KSCCs achieve 2.3–2.4× speedup. The *filter ratio* $|C|/|\Gamma(W)|$ is the direct predictor of speedup, not KSCC% alone. KSCC concentration contributes to a low filter ratio, but graph density moderates the realised benefit: soc-Epinions1 has the largest KSCC (42.5%) but a higher filter ratio (0.643) due to its large absolute $\Gamma(W)$.

G. Experiment 7: Edge Category Analysis

On Wiki-Vote, SG-SNIR selects edges in exactly the same structural categories as exhaustive MaxExpectedH: 50% internal KSCC edges, 16.7% bridge edges, and 33.3% peripheral edges. This confirms that the SpectralDrop filter identifies the same structurally critical edges as exhaustive simulation, at 2.29× lower evaluation cost. On HIV, both methods select 100% peripheral edges, consistent with the KSCC containing no $\Gamma(W)$ candidates.

H. Experiment 8: Robustness to Seed Configuration

Table VIII reports the stability of SG-SNIR across varying seed counts ($k = 10$, 10 trials per seed size) on two contrasting datasets: HIV (heterogeneous degree distribution) and p2p-Gnutella (structured KSCC).

TABLE VIII
ROBUSTNESS TO SEED CONFIGURATION: MEAN H (COV%).

Seeds	HIV		p2p-Gnutella	
	Mean	CoV	Mean	CoV
1	4.25	32.9%	3.13	0.0%
3	15.44	27.4%	13.37	4.5%
5	39.21	68.0%	24.12	9.6%
10	81.24	34.3%	50.04	5.9%
20	169.17	21.8%	102.40	3.3%

p2p-Gnutella shows high stability (CoV $\leq$ 10% across all seed sizes; deterministic at seeds=1). HIV shows high variance, peaking at CoV=68% for seeds=5. This reflects HIV's heterogeneous degree distribution: seed configurations hitting hub nodes produce outbreaks up to 7× larger than peripheral seeds. This is a property of the *network*, not the algorithm. All inter-method comparisons in this work use identical seeds (BASE_SEED=42) to ensure fair evaluation.

I. Experiment 9: Comparison with State-of-the-Art Baselines

To validate the necessity of SG-SNIR's hybrid approach, we benchmark it against the two core frameworks it builds upon: the exhaustive simulation strategy introduced for the SNIR model in [1] (which we implement as **MaxExpectedH**), and the purely structural spectral node-immunization method, **DINO** [2].

Comparison against Paper 1 [1]: As established in Table III and Table IV, SG-SNIR matches the quality of the MaxExpectedH approach (from [1]) within 0.1–2.3% while achieving up to a 2.4× speedup. This proves that our spectral filtering safely prunes the search space, overcoming the computational intractability of [1]'s naive greedy search without sacrificing epidemic realism. Furthermore, Table VI demonstrates that MaxExpectedH fails to scale on networks exceeding 200K nodes (email-EuAll), where SG-SNIR remains fully tractable.

Comparison against DINO [2]: DINO [2] is a node immunization strategy that greedily removes nodes to minimize the spectral radius of the KSCC. However, removing whole nodes is often practically impossible in real-world quarantines. To create a fair comparison against SG-SNIR's targeted edge-blocking approach, we evaluate the interventions based on *Network Disruption*—the equivalent contact budget measured by the total number of connections severed. Node removal is a coarse-grained intervention that inflicts massive structural damage to the network, whereas targeted edge removal is fine-grained.

We subjected both algorithms to massive, globally distributed epidemics by initializing the SNIR simulation with 300 and 600 random seed nodes in the p2p-Gnutella network. We use large seed counts (300, 600) for this comparison to create outbreak conditions where DINO's node removal strategy selects meaningful targets — with only 5 seeds, the KSCC containment effect is too small to differentiate the methods clearly. We ran DINO to incrementally select varying budgets of top hub nodes to quarantine. For 300 seeds, quarantining up to 15 central nodes severed exactly 1,256 edges (the sum of in-degree plus out-degree of the 15 selected nodes in the original graph). For 600 seeds, quarantining up to 20 nodes severed 1,607 edges. We then provided SG-SNIR with the exact same maximum edge-disruption budgets and compared the resulting expected final influence H_{final}.

As shown in Fig. 3, SG-SNIR vastly outperforms DINO across all scales of network disruption. Under the massive 300-seed outbreak, DINO's maximal disruption (severing 1,256 edges) only reduced the spread to $H_{\text{final}} = 1136.50$. When granted the exact same disruption budget, SG-SNIR successfully isolated the critical transmission vectors, achieving a much lower $H_{\text{final}} = 938.82$. Similarly, under the extreme 600-seed outbreak, SG-SNIR achieved $H_{\text{final}} = 1910.60$ compared to DINO's 2246.54 for the same 1,607-edge disruption cost. This conclusively validates our core argument: structural metrics alone waste intervention resources breaking non-transmitting contacts. SG-SNIR leverages spectral guidance combined with SNIR-awareness to effectively trap the

epidemic while preserving the underlying integrity of the network.

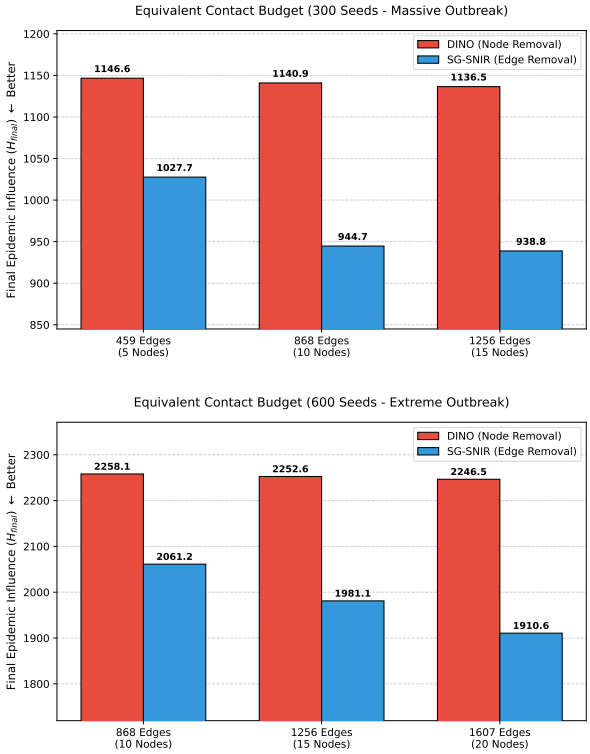

Fig. 3. Epidemic containment under equivalent Network Disruption budgets for 300-seed (top) and 600-seed (bottom) massive outbreaks on p2p-Gnutella. SG-SNIR consistently achieves significantly lower H_{final} than DINO at all budget levels, effectively halting the epidemic while inflicting the exact same amount of structural damage to the network.

VIII. Conclusions

We presented SG-SNIR, a hybrid spectral-guided framework for epidemic containment via edge blocking in directed networks. By combining KSCC-based structural filtering with an $\mathcal{O}(1)$-per-edge SpectralDrop criterion and full SNIR simulation on the reduced candidate set, SG-SNIR achieves a favourable quality-efficiency trade-off: 2.3–2.4$\times$ fewer epidemic simulations at a quality cost of 0.1–2.3% compared to exhaustive MaxExpectedH on networks with meaningful KSCC structure. On networks without KSCC structure (HIV), SG-SNIR degrades gracefully to MaxExpectedH with zero quality loss.

The spectral threshold ε is robust across two orders of magnitude, attributed to the tightly concentrated SpectralDrop distribution on real networks. The ablation study confirms that every component — KSCC filtering, bridge edge inclusion, and spectral thresholding — contributes to efficiency without sacrificing quality.

Future work includes supporting dynamic recalculation of W as the epidemic evolves, extending the framework to heterogeneous transmission probabilities, and scaling to billion-edge networks through approximate SCC decomposition.

References

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